

```
Processando config 28, seed 3, teste: BlockAlphabit
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 2
Value of the statistic : 1.63
p-value of test : 0.44
```

```
-----
CPU time used : 00:00:00.03
```

```
Generator state:
16777216 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 10.09  
p-value of test : 0.26

-----  
CPU time used : 00:00:00.03

Generator state:

33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 114.44  
p-value of test : 0.80

-----  
CPU time used : 00:00:00.03

Generator state:

50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32659.53  
p-value of test : 0.66

-----  
CPU time used : 00:00:00.03

Generator state:

67108864 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 139.37  
p-value of test : 0.98

-----  
CPU time used : 00:00:00.02

Generator state:

83886080 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 295.80  
p-value of test : 0.64

-----  
CPU time used : 00:00:00.02

Generator state:

100663296 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : -0.33  
p-value of test : 0.63

-----  
CPU time used : 00:00:00.03

Generator state:

117440512 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 40.76 |
| p-value of test              | : | 0.14  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 29.05 |
| p-value of test              | : | 0.57  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 29.30 |
| p-value of test              | : | 0.60  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 22.07 |
| p-value of test              | : | 0.63  |

-----  
Test on the values of the Statistic C

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 15     |
| ChiSquare statistic          | : | 4.89   |
| p-value of test              | : | 0.9930 |

-----  
CPU time used : 00:00:00.06

Generator state:

134217728 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 71.14 |
| p-value of test              | : | 0.20  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 59.97 |
| p-value of test              | : | 0.55  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 172.76 |
| p-value of test              | : | 0.23   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 42.88 |
| p-value of test              | : | 0.92  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 31.22 |
| p-value of test              | : | 0.51  |

-----  
CPU time used : 00:00:00.06

Generator state:

150994944 bits have been read.

===== Summary results of Alphabit =====

|                       |                                                |
|-----------------------|------------------------------------------------|
| Version:              | TestU01 1.2.3                                  |
| Generator:            | ufile_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits:       | 16777216                                       |
| Number of statistics: | 17                                             |
| Total CPU time:       | 00:00:00.33                                    |

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
```

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----

Test Results for Delta = 1.0000

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 2    |
| Value of the statistic       | : | 1.33 |
| p-value of test              | : | 0.52 |

-----

CPU time used : 00:00:00.03

Generator state:

167772160 bits have been read.

```
*****
```

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 8    |
| Value of the statistic       | : | 2.26 |
| p-value of test              | : | 0.97 |

-----  
CPU time used : 00:00:00.02

Generator state:

184549376 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216

Number of cells =  $2^L$  = 256

Expected number per cell = 65536

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 128    |
| Value of the statistic       | : | 106.01 |
| p-value of test              | : | 0.92   |

-----  
CPU time used : 00:00:00.00

Generator state:

201326592 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 16,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 0.00097653271,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 32768
Value of the statistic            : 32581.08
p-value of test                   : 0.77
```

```
-----
CPU time used                     : 00:00:00.03
```

Generator state:

218103808 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

sstring\_HammingIndep test:

```
-----
N = 1,  n = 524288,  r = 0,  s = 32,  L = 16,  d = 0
```

Counters with expected numbers >= 10

```
-----
Number of degrees of freedom      : 177
Chi-square statistic             : 146.28
p-value of test                   : 0.96
```

```
-----
CPU time used                     : 00:00:00.03
```

Generator state:

234881024 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 305    |
| Chi-square statistic         | : | 331.62 |
| p-value of test              | : | 0.14   |

-----

|               |   |             |
|---------------|---|-------------|
| CPU time used | : | 00:00:00.00 |
|---------------|---|-------------|

Generator state:

251658240 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----

|                  |   |       |
|------------------|---|-------|
| Normal statistic | : | -1.89 |
| p-value of test  | : | 0.97  |

-----

|               |   |             |
|---------------|---|-------------|
| CPU time used | : | 00:00:00.02 |
|---------------|---|-------------|

Generator state:

268435456 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 27.41 |
| p-value of test              | : | 0.70  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 47.24 |
| p-value of test              | : | 0.03  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 24.26 |
| p-value of test              | : | 0.83  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 41.75 |
| p-value of test              | : | 0.02  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 22.06 |
| p-value of test              | : | 0.11  |

-----  
CPU time used : 00:00:00.06

Generator state:

285212672 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 52.10 |
| p-value of test              | : | 0.81  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 51.06 |
| p-value of test              | : | 0.84  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 204.64 |
| p-value of test              | : | 9.9e-3 |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 46.07 |
| p-value of test              | : | 0.85  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 23.30 |
| p-value of test              | : | 0.87  |

-----  
CPU time used : 00:00:00.05

Generator state:

301989888 bits have been read.

===== Summary results of Alhabit =====

|                       |                                                |
|-----------------------|------------------------------------------------|
| Version:              | TestU01 1.2.3                                  |
| Generator:            | ufire_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits:       | 16777216                                       |
| Number of statistics: | 17                                             |
| Total CPU time:       | 00:00:00.28                                    |

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 0.72
p-value of test                   : 0.70
```

```
-----
CPU time used                     : 00:00:00.02
```

Generator state:

318767104 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
```

Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For  $\Delta > -1$ , we use the ChiSquare approximation  
Correction factor of the ChiSquare:

$\Delta = 1$ ,  $\mu = 2.0861626e-07$ ,  $\sigma = 1$

-----  
Test Results for  $\Delta = 1.0000$

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 8      |
| Value of the statistic       | : | 1.61   |
| p-value of test              | : | 0.9908 |

-----  
CPU time used : 00:00:00.03

Generator state:  
335544320 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin\_MultinomialBitsOver test:

-----  
 $N = 1$ ,  $n = 16777216$ ,  $r = 0$ ,  $s = 32$ ,  $L = 8$ ,  
Sparse = FALSE

Number of bits =  $n = 16777216$   
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For  $\Delta > -1$ , we use the ChiSquare approximation  
Correction factor of the ChiSquare:

$\Delta = 1$ ,  $\mu = 3.784895e-06$ ,  $\sigma = 1$

-----  
Test Results for  $\Delta = 1.0000$

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 128    |
| Value of the statistic       | : | 135.41 |
| p-value of test              | : | 0.31   |

-----  
CPU time used : 00:00:00.02

Generator state:  
352321536 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32847.27  
p-value of test : 0.38

-----  
CPU time used : 00:00:00.03

Generator state:

369098752 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 210.57  
p-value of test : 0.04

-----  
CPU time used : 00:00:00.02

Generator state:

385875968 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 308.47  
p-value of test : 0.43  
-----

CPU time used : 00:00:00.02

Generator state:

402653184 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 0.57  
p-value of test : 0.28  
-----

CPU time used : 00:00:00.02

Generator state:

419430400 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 21.56 |
| p-value of test              | : | 0.92  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 25.52 |
| p-value of test              | : | 0.74  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 36.70 |
| p-value of test              | : | 0.26  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 14.24 |
| p-value of test              | : | 0.96  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 12.15 |
| p-value of test              | : | 0.67  |

-----  
CPU time used : 00:00:00.06

Generator state:

436207616 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 60.13 |
| p-value of test              | : | 0.54  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 48.14 |
| p-value of test              | : | 0.90  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 140.23 |
| p-value of test              | : | 0.87   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 60.15 |
| p-value of test              | : | 0.36  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 40.46 |
| p-value of test              | : | 0.14  |

-----  
CPU time used : 00:00:00.05

Generator state:

452984576 bits have been read.

===== Summary results of Alphabit =====

|                 |                                                |
|-----------------|------------------------------------------------|
| Version:        | TestU01 1.2.3                                  |
| Generator:      | ufile_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits: | 16777216                                       |

Number of statistics: 17  
Total CPU time: 00:00:00.27

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alhabit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 2
Value of the statistic : 6.78
p-value of test : 0.03
```

```
-----
CPU time used : 00:00:00.02
```

Generator state:  
469761792 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

-----  
Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic       : 10.09
p-value of test              : 0.26
```

-----  
CPU time used : 00:00:00.02

Generator state:

486539008 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

-----  
Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic       : 148.69
p-value of test              : 0.10
```

-----  
CPU time used : 00:00:00.02

Generator state:

503316224 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 33125.41  
p-value of test : 0.08

-----  
CPU time used : 00:00:00.03

Generator state:

520093440 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 164.55  
p-value of test : 0.74

-----  
CPU time used : 00:00:00.02

Generator state:  
536870656 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 305    |
| Chi-square statistic         | : | 302.95 |
| p-value of test              | : | 0.52   |

-----  
CPU time used : 00:00:00.00

Generator state:  
553647872 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 1.03  
p-value of test : 0.15

-----  
CPU time used : 00:00:00.02

Generator state:  
570425088 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk\_RandomWalk1 test:

```
-----
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64
```

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 50.55 |
| p-value of test              | : | 0.02  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 35.63 |
| p-value of test              | : | 0.26  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 31.18 |
| p-value of test              | : | 0.51  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 15.56 |
| p-value of test              | : | 0.93  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 16.62 |
| p-value of test              | : | 0.34  |

-----  
CPU time used : 00:00:00.06

Generator state:

587202304 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config28_seed3.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 59.44 |
| p-value of test              | : | 0.57  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 63.96 |
| p-value of test              | : | 0.41  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 142.97 |
| p-value of test              | : | 0.83   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 56.09 |
| p-value of test              | : | 0.51  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 27.38 |
| p-value of test              | : | 0.70  |

-----  
CPU time used : 00:00:00.06

Generator state:  
603979264 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3

Generator: ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits: 16777216  
Number of statistics: 17  
Total CPU time: 00:00:00.25

All tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alphabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 2  
Value of the statistic : 2.42  
p-value of test : 0.30

-----  
CPU time used : 00:00:00.02

Generator state:  
620756480 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 8    |
| Value of the statistic       | : | 3.91 |
| p-value of test              | : | 0.87 |

-----  
CPU time used : 00:00:00.02

Generator state:  
637533696 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 128    |
| Value of the statistic       | : | 123.68 |
| p-value of test              | : | 0.59   |

-----  
CPU time used : 00:00:00.00

Generator state:  
654310912 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----

Test Results for Delta = 1.0000

|                              |            |
|------------------------------|------------|
| Number of degrees of freedom | : 32768    |
| Value of the statistic       | : 32819.54 |
| p-value of test              | : 0.42     |

-----

CPU time used : 00:00:00.03

Generator state:

671088128 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----

|                              |          |
|------------------------------|----------|
| Number of degrees of freedom | : 177    |
| Chi-square statistic         | : 172.35 |
| p-value of test              | : 0.58   |

-----

CPU time used : 00:00:00.02

Generator state:  
687865344 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 305    |
| Chi-square statistic         | : | 315.41 |
| p-value of test              | : | 0.33   |

-----

CPU time used : 00:00:00.00

Generator state:  
704642560 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----

|                  |   |       |
|------------------|---|-------|
| Normal statistic | : | -0.22 |
| p-value of test  | : | 0.59  |

-----

CPU time used : 00:00:00.02

Generator state:  
721419776 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 29.65 |
| p-value of test              | : | 0.59  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 22.88 |
| p-value of test              | : | 0.85  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 26.47 |
| p-value of test              | : | 0.74  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 24.08 |
| p-value of test              | : | 0.51  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 19.69 |
| p-value of test              | : | 0.18  |

-----  
CPU time used : 00:00:00.06

Generator state:

738196992 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 47.77 |
| p-value of test              | : | 0.91  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 56.35 |
| p-value of test              | : | 0.68  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 170.98 |
| p-value of test              | : | 0.26   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 48.06 |
| p-value of test              | : | 0.79  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 31.62 |
| p-value of test              | : | 0.49  |

-----  
CPU time used : 00:00:00.05

Generator state:  
754973952 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3  
Generator: ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits: 16777216  
Number of statistics: 17  
Total CPU time: 00:00:00.25

All tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alhabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 2  
Value of the statistic : 1.27  
p-value of test : 0.53

-----

CPU time used : 00:00:00.02

Generator state:

771751168 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 10.85  
p-value of test : 0.21

-----  
CPU time used : 00:00:00.02

Generator state:

788528384 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 108.86  
p-value of test : 0.89

-----  
CPU time used : 00:00:00.00

Generator state:  
805305600 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32688.41  
p-value of test : 0.62

-----

CPU time used : 00:00:00.03

Generator state:  
822082816 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 177  
Chi-square statistic : 184.50  
p-value of test : 0.33

-----  
CPU time used : 00:00:00.02

Generator state:  
838860032 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 329.96  
p-value of test : 0.16

-----  
CPU time used : 00:00:00.00

Generator state:  
855637248 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 1.65  
p-value of test : 0.05

-----  
CPU time used : 00:00:00.02

Generator state:  
872414464 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 42.08 |
| p-value of test              | : | 0.11  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 29.53 |
| p-value of test              | : | 0.54  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 23.53 |
| p-value of test              | : | 0.86  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 18.19 |
| p-value of test              | : | 0.83  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 12.99 |
| p-value of test              | : | 0.60  |

-----  
CPU time used : 00:00:00.06

Generator state:

889191680 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config28\_seed3.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 56.70 |
| p-value of test              | : | 0.67  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 66.74 |
| p-value of test              | : | 0.32  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 148.06 |
| p-value of test              | : | 0.74   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 52.58 |
| p-value of test              | : | 0.64  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 22.62 |
| p-value of test              | : | 0.89  |

-----  
CPU time used : 00:00:00.05

Generator state:

905968640 bits have been read.

===== Summary results of Alphabit =====

Version:            TestU01 1.2.3  
Generator:          ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits:     16777216  
Number of statistics: 17  
Total CPU time:     00:00:00.25

All tests were passed